MIL-F-13785A
30 NOVEMBER 1955
SUPERSEDING
MIL-F-13785 (Cm1C)
12 NOVEMBER 1954

MILITARY SPECIFICATION

FILTER MATERIAL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers filter material for use in removing particulates from influent air in gas mask canisters and collective protectors.

1.2 Classification. The filter material shall be furnished in the following types as specified (see 6.1):

Type 5 — Abestos-bearing (low resistance).

Type 6 — Asbestos-bearing (high resistance).

Type 7 — Asbestos-bearing (high resistance).

Type 8 — Glass-fiber-bearing (high resistance).

Type 15-N — Asbestos and/or glass bearing (low resistance).

Types 6 and 7 differ in composition (see table I). Types 7 and 8 are intended for use only in M11-type canister filters conforming to Specification MIL-F-10969. Type 15-N is intended for use in Navy C-1 cannisters.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

O-P-94	Paper, Test; pH Indicator.
NN-B-621	Boxes, Wood, Nailed and Lock-Corner.
UU-P-31	Paper; General Specifications and Methods of Testing.
CCC-T-191	Texile Test Methods.

MILITARY

JAN-P-125	Packaging and Packing for Overseas Shipment —Barrier-Materials, Waterproof, Flexible.
JAN-P-127	Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water Resistant.
MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier-Material.
MIL-F-10462	Filter Units, Protective Clothing, Gas Mask

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:19Z
Check the source to verify that this is the current version before use.

Components and Related Products; Performance Test Methods.

MIL–F–10969 Filter, Element, Aerosol.

STANDARDS

MILITARY

MIL–STD–129 Marking for Shipment and Storage.

PUBLICATIONS

CHEMICAL CORPS DIRECTIVE

334 Operation of Apparatus, Canister Testing, MIT–E3.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The furnish for the filter sheet shall be made with suitable proportions of the fibers shown in table I.

TABLE 1. *Composition*

Type	*Fibers*[1, 2]
5	b, r, f, h
6	b, e, f, h
7	b, r, a (or i) h
8	g, r, a (or i) h
15-N	b (or g) f, h

[1] b—asbestos (Bolivan crocidolite) see 6.2
r—viscose rayon
f—cotton flock
h—manila hemp
e—esparto
a—hardwood sulfate pulp
i—hardwood sulfite pulp
g—glass fibers (0.4 to 0.6 micron diameter)

[2] Fibers other than those listed may be used only after test and approval by the Chemical Corps.

3.1.1 *Gauze backing.* Gauze backing is required for type 5 material (see 6.3). The gauze backing shall be treated with at least 2 percent of mildew inhibitor based on the dry weight of the gauze (see 4.4.8 and 6.4).

3.2 **Physical properties.** The filter material shall conform to table II when tested as specified in 4.4.

3.3 **Workmanship.** The filter material shall contain no thick or thin spots, abrasions, tears, punctures, wrinkles, dirt inclusions, or other imperfections which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Lot.** A lot shall consist of the filter material manufactured on one machine under substantially the same conditions in not more than 24 consecutive hours. The inspection lot for nondestructive inspection within the above limitations shall be designated by the inspector.

4.2 **Sampling.** A sample 6 feet long for type 5 and 3 feet long for types 6, 7, 8, and 15–N material shall be taken from the free end of each slit roll across the entire width of the roll, a sufficient distance from the end to prevent inclusion of material damage due to shipping and wrapping. Each sample shall be identified with the roll from which it was taken (see 6.7).

4.3 **Inspection.**

4.3.1 *Visual.* Each sample shall be visually examined for compliance with 3.3 before the test strips are prepared.

4.3.2 *Winding, packaging, packing, and marking.* The inspector shall inspect the winding, packaging, packing, and marking of the filter material for compliance with section 5.

4.4 **Tests.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:19Z
Check the source to verify that this is the current version before use.

TABLE II. *Physical Requirements*

Type	5	6, 7, 8	15-N
Width	As specified (see 3.1.3 and 6.1)		
Thickness (inch)	(Min.) (Max.) 0.019 0.030	(Min.) (Max.) 0.033 0.043	(Min.) (Max.) 0.035 0.050
Air resistance in mm. water (flowrate—32 l./min.)	19 27	— 44	— 27
(Layers tested)	(5)	(1)	(1)
DOP penetration (%) Flowrate 32 l./min.	— 0.30	— 0.05	— 0.026
(Layers tested)	(5)	(1)	(1)
Tensile strength lb./inch width	not required	2.0 —	1.0 —
Water repellency Water column to be supported without immediate penetration (see 6.6) (inches)	15 —	20 —	20 —
(Layers tested)	(7)	(1)	(1)
Acidity	To pass test		
Mildew inhibitor (%) (see 4.4.8 and 6.4)	2.5 of dry weight of furnish		
Fold characteristics	Not required	To pass test	Not required

4.4.1 *Test panels.* The sample shall be divided into test panels, the length of which in the machine direction shall be as shown in table III and the width shall be the width of the roll. Test strips shall be cut from the test panels at equally spaced intervals. The number of test strips shall be equal to the width of the roll in inches divided by 10 plus one sample for the reminder. For example, a 42-inch roll would require 5 test strips, therefore the test panel should be cut into 5 approximately 8-inch strips. If the roll is less than 10 inches in width, the test strip width shall be the width of the roll. Each panel shall be subjected to the tests indicated in table III.

TABLE III. *Length of test panels*

Test	Type 5	Types 6, 7, 8, and 15-N
DOP penetration and air resistance	28 inches	7 inches
Thickness	3 inches	3 inches
Tensile strength	8 inches (min.)	8 inches (min.)
Fold characteristics	.....	6 inches
Water repellency	21 inches	3 inches

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:19Z
Check the source to verify that this is the current version before use.

4.4.2 *Width.* Each roll shall be checked for compliance with the width specified.

4.4.3 *Thickness.*

4.4.3.1 *Gage.* The gage shall be of the dead-weight type equipped with a dial graduated to read directly to 0.001 inch. The presser foot shall be circular with a diameter of 0.25 ± 0.01 inch. The presser foot and moving parts connected therewith shall be weighted so as to apply a total lead of 3.0 ± 0.1 ounce to the specimen. The presser foot and anvil shall be plane and parallel to each other.

4.4.3.2 *Procedure.* The specimen, cut to a size convenient to handle, shall be placed upon the anvil of the gage, smooth but without tension. The presser foot shall be lowered upon it gradually, allowed to come to rest without sharp impact and the reading of the dial then observed. The thickness of the specimen at each of at least three places measured shall be within the applicable range specified.

4.4.4 *DOP penetration and air resistance.* The DOP penetration and air resistance shall be determined on each test pad, approximately 5 inches square (5 plies for type 5 and 1 ply for types 6, 7, 8, and 15–N material), using the method specified in Directive 334 and as follows: Insert the test pad into an adapter, cutting the test pad as necessary to fit the adapter (see 6.8). Insert the adapter containing the test pad in the apparatus in place of the canister holder and draw DOP aerosol through an area of 100 sq. cm. (4.443-inch diameter round test pad) at a flowrate of 32 liters per minute. Read the DOP aerosol penetration on the meter scale and the air resistance on a water manometer connected to the adapter holding the test pad.

4.4.5 *Tensile strength.* The tensile strength shall be determined as specified in method 171 of Specification UU–P–31.
In addition:

(a) The machine shall be calibrated to be read accurately, in the region of the test results, in 0.1-pound increments.

(b) Tests shall be on strips cut in the paper-machine direction only.

(c) Test specimens shall be not less than 1 inch nor greater than 2 inches wide.

(d) The length of the test specimens under stress shall be not less than 6 inches.

(e) The result of tests shall be reported as strength in pounds per inch width of specimen.

(f) The test specimens shall be conditioned and tested as specified in Method 102 of Specification UU–P–31.

4.4.6 *Water repellency.* The test pads shall consist of 7 plies for type 5 material and 1 ply for types 6, 7, 8, and 15–N material. The test pads shall be cut as necessary to fit the aparatus specified in Method 603.1 of Specification MIL–F–10462 and the water repellency shall be determined as specified in that method. The results shall be checked for compliance with the applicable portion of table II.

4.4.7 *Acidity.* The pH of each roll of paper shall be determined immediately after treatment for water repellency as follows: Moisten a strip of type I, class B, indicator paper conforming to Specification O–P–94 with distilled water. Place the moist indicator paper flat on the outer layer of the roll of paper. Within 10 seconds, check the indicator paper for color change. The color shall be as specified in Specification O–P–94 for a pH of 7.0 — 8.0. If the color does not indicate a pH of 7.0 — 8.0 the roll represented shall be rejected. Rejected rolls may be given an additional ammonia treatment (but no further water-repellency treatment) on the

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:19Z
Check the source to verify that this is the current version before use.

waterproofing machine and then resubmitted for acidity test.

4.4.8 *Mildew inhibitor.*

4.4.8.1 *Certification.* The contractor shall furnish the inspector with a certificate giving the name, percentage, and manufacturer of the mildew inhibitor used as specified in 3.1.1 and table II. However, the contracting officer, at his discretion, may require that the gauze or the filter material be tested as specified in 4.4.8.2.

4.4.8.2 *Test.* Test specimens of the material as specified in Method 5751 of Specification CCC-T-191. Evaluation shall be qualitative (the specimens shall show no growth when compared with the controls at the end of the test period).

4.4.9 *Fold characteristics.* Each test strip, cut 6 inches long in the machine direction and 3 inches wide, shall be bent across the machine direction through an arc of 180° over a 3/16-inch-diameter round mandrel. No sample shall tear, break, or crack while being tested. Superficial surface separation shall not be cause for rejection.

4.5 Rejection and retest. If any roll fails to comply with the specified width or the requirements of 3.3, the roll shall be rejected. The lot shall be rejected if any sample of filter material or gauze backing fails to meet the mildew inhibitor requirement when tested. If any sample fails to meet the requirements for tensile strength, thickness, water repellency, DOP penetration, acidity, fold characteristics, or air resistance as specified in table II the sample shall be retested in not less than 24 hours. In the case of tensile strength failure new test strips shall be cut from the test panels. If the sample fails to meet the requirements when retested, the roll represented and the succeeding roll in the lot shall be rejected (see 6.5). Manufacturers may resubmit the material for acceptance after removal of the defective portions of the material.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A, military package.* Not applicable.

5.1.2 *Level B, limited military package.* The filter material shall be wound evenly on a hollow core, constructed of paper of other suitable material, having an inside diameter of 3-1/16 ± ⅛ inches. A maximum of four splices is permitted for each roll. The location of the splices shall be indicated by a paper "flag" having a contrasting color. Each roll shall have a maximum outside diameter of 25 inches and not more than 3 percent of the rolls in a lot shall have an outside diameter of less than 20 inches. No splices shall be permitted on the roll in that portion exceeding 20 inches in diameter. The rolls shall be wrapped with two thicknesses of waterproof barrier material conforming to type C-1 of Specification JAN-P-125. The wrapper shall be closed at the ends by means of inside and outside headers, using sufficient adhesive to make watertight seals. The barrier material shall be sealed with an adhesive conforming to type II, grade B of Specification MIL-A-140.

5.1.3 *Level C, minimum military package.* Package as specified in 5.1.2.

5.2 Packing.

5.2.1 *Level A, military pack.* Not applicable.

5.2.2 *Level B, limited military pack.* The filter material, packaged as specified in 5.1.2, shall be packed in wood boxes conforming to style 4 of Specification NN-B-621. The net weight of the contents of each box shall not exceed 250 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:19Z
Check the source to verify that this is the current version before use.

5.2.3 *Level C, minimum military pack.* Pack as specified in 5.2.2.

5.3 **Marking.**

5.3.1 *Roll identification.* Each slit roll shall be marked with the lot number, and the reel number and the location in the reel from which it was taken in the following manner:

(a) Each reel in the lot shall be marked consecutively starting with number 1. All rolls rewound from one reel shall have the same number as the reel.

(b) Each roll rewound from one reel shall be marked alphabetically, with the letter A indicating the first roll rewound from the reel. Successive rolls from the same reel shall be marked B, C, etc.

(c) In addition, slit rolls shall be numbered consecutively, with number 1 indicating the "front position" on the machine. (The "front position" is that side of the machine from which the machine tender controls the sheet as it progresses toward the reel.

5.3.2 *Shipments.* In addition to any special marking required by the contract or order, each shipping container shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify:

(a) Title, number and date of this specification.

(b) Type of filter material required.

(c) Width of rolls to be supplied.

(d) In addition, procurement documents should require the contractor to certify the amount and type (company and trade designation) of the mildew inhibitor used.

6.2 **Asbestos stock pile.** Specially screened asbestes fiber suitable for the manufacture of asbestos bearing filter material may be obtained from a stock pile maintained by the Chemical Corps at the Army Chemical Center, Maryland.

6.3 **Gauze backing.** The thickness requirement for type 5 material has been met by using gauze backing having 12–20 threads per inch in the warp and 11–18 in the fill. However, any other gauze may be used which will enable the finished paper to meet the thickness requirement of this specification.

6.4 **Mildew inhibitors.** The following mildew inhibitors have been found satisfactory:

Shirlan A

Zerlate (Milban D)

If other materials are proposed for use, they should be approved by the Office of the Chief Chemical Officer, before use.

6.5 **Roll samples.** The sample selected represents not only the end of the roll being sampled but also the start of the succeeding roll and therefore in case of a failure the succeeding roll should also be rejected.

6.6 **Water repellency.** Satisfactory water-repellent filter material has been prepared by the Chemical Corps by treating the material with methyl silicon chloride vapors in accordance with Chemical Corps Directive 233. Manufacture of Water Repellent Filter Material. Use of this directive is not mandatory nor does its use constitute a guarantee of acceptance of material so treated. Any process imparting water repellency in accordance with this specification may be used, provided approval of the process is obtained from the Office of the Chief Chemical Officer.

6.7 **Sampling and testing.** When the contractor consistently produces high-quality

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:19Z
Check the source to verify that this is the current version before use.

material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.8 Adapter. A satisfactory adapter is shown on Chemical Corps Drawings B21-52-143 and B21-52-145. The use of this particular adapter is not mandatory, however.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Chemical Corps
Navy—Bureau of Ships

Other interests:
Army—M
Navy—Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:19Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-6278

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:19Z
Check the source to verify that this is the current version before use.